Sahaj Bhadja

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Summary

Machine Learning Engineer with 2+ years building **LLM/NLP systems**, **model evaluation** pipelines, Python services, large-data workflows, and production-ready AI applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built **LLM orchestration** platform with FastAPI jobs, explicit states, failure isolation, and reusable status APIs, processing 1K+ AI tasks while tripling throughput for concurrent workflows and dashboards.
- Designed **SQL-backed** data platform with normalized schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built **multimodal LLM** grading workflows with React, Flask, OCR, and model selection, supporting 150+ instructors and 10K+ daily requests through reliable production dashboards and APIs during peaks.
- Designed **evaluation pipelines** for rubric generation, feedback, token tracking, and secure file ingestion, reducing grading turnaround 60% while keeping batch processing observable during peak instructor submissions reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built **Decision Transformer** pipelines for autonomous-driving offline RL, improving planning 18% in simulation while processing 100K+ trajectory logs through reproducible PyTorch evaluation workflows for research experiments repeatedly.
- Architected **spatial-temporal encoder** with HD-map geometry, lane context, ego trajectories, and multi-head attention, improving closed-loop stability 15% across simulated baselines for ML research workflows and evaluation reports.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Engineered **production CI/CD** pipelines for cross-platform microservices with Jenkins, VM automation, and 20+ PyTest, Mockito, unit, and integration suites across Agile release cycles and QA validation reviews.
- Troubleshot **large-scale data** integrations using OOP patterns, SQL workflows, Amazon Web Services Glue, and S3, improving response times 30% while reducing technical debt 20% across production releases.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship *India*
- Benchmarked **machine learning** classifiers including Naive Bayes, Random Forest, RNN, LSTM, Bi-LSTM, GRU, and CNN models for malicious URL detection and comparative analysis experiments reproducibly with metrics.
- Prepared **model evaluation** reports across lexical URL features, sequence models, and classification baselines, strengthening debugging discipline for false positives, false negatives, and reproducible experiments iteratively with documentation.

Skills

Languages & Math: Python, Java, C++, JavaScript, TypeScript, SQL, Data Structures, Algorithms, Linear Algebra, Probability, Statistics, OOP
ML, NLP & LLMs: PyTorch, Transformers, LLM Workflows, Prompting, OCR, NLP, NLU, Summarization, Semantic Search, Embeddings, Intent Recognition, Dialogue Evaluation, Model Evaluation, Offline RL, CNNs, RNNs, LSTM, GRU
Data & Production: FastAPI, Flask, REST APIs, PostgreSQL, SQL Server, MySQL, MongoDB, AWS Glue, S3, Large Datasets, Labeling Workflows, Data Quality Checks, Monitoring, Debugging
Delivery & Research: CI/CD, Jenkins, GitHub Actions, Docker, Git, Unit Testing, Integration Testing, PyTest, Experiment Design, Benchmarking, Documentation, Agile, Model Governance

Projects

Voice Interaction Evaluation Harness | *Python, Transformers, FastAPI, SQL, LLM Workflows, Evaluation*
- Designed **voice-agent evaluation** concept for transcript labeling, intent checks, hallucination review, latency buckets, structured outputs, and SQL-backed validation reports for regulated call-quality workflows and audit reviews safely.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*
- Built **semantic retrieval** over 50K+ documents using transformer embeddings and FAISS, enabling contextual Q&A, summarization, and sub-second retrieval through optimized similarity pipelines for users online reliably at scale.

CaptionCraft Live | *Transformers, PyTorch, Flask, Streamlit, Gradio*
- Benchmarked **transformer models** including BLIP, GIT, ViT-GPT-2, and custom captioning on 10K images and 100K captions, improving caption accuracy while building scalable inference APIs for users online.